

**Why Don't You Just Meet Me in the Middle:
Intrahousehold Bargaining and Divorce Reform in Mexico**

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Abstract

Lowering barriers to divorce matters because it not only lets spouses leave bad marriages, but also the threat of exit itself can trigger renegotiation of the terms of marriages that continue. This paper examines how Mexico's adoption of no-fault unilateral divorce (NFUD) reallocated consumption within married households. On average across households, NFUD increased the share of the household budget spent on children by 5.2 points per child by shifting spending away from both parents. While prior work has framed bargaining gains as flowing to either men or women, I find the reform gave the spouse more likely to file, regardless of gender, leverage to direct more of the couple's resources toward children rather than personal consumption. The shift is largest in couples where unilateral divorce was more likely to occur. These findings shed new light on the intrahousehold bargaining dynamics that determine spending on children.

1 Introduction

While previous research has documented how the liberalization of divorce laws affects a broad range of outcomes for women (e.g., divorce rates, intimate partner violence, labor force participation), including for those who remain married, identifying the causal link between the two typically requires making significant assumptions about the underlying mechanism at work. For example, a study showing that a decrease in the cost of divorce lowers incidence of intimate partner violence might argue that this relationship is due to an improvement in women's outcome post-divorce and consequently her ability to credibly threaten to leave if a marriage becomes or continues to be abusive. Yet the value of her outside option and the credibility of her threat are not directly observed; often this underlying relationship is assumed in theory (Jayachandran & Voena, 2026). Rather than assuming that an outcome shows that women's bargaining power increased or decreased, I use a household bargaining model to derive testable predictions about how Mexico's adoption of no-fault unilateral divorce (NFUD) would affect intrahousehold resource shares across couples if power within the household had shifted, thereby converting the identifying assumption into a testable pattern.

Intrahousehold resource shares, defined as the share of total household spending on each member, are a leading metric of inequality within the household since they are based on a meaningful welfare outcome that is comparable across institutional contexts. Yet resource shares have been mostly used to describe intrahousehold inequality rather than to evaluate the effects of policy interventions (e.g., Calvi et al., 2023). The main challenge is that shares are not observed directly from household expenditure data; without information on actual consumption as well as individual preferences for that consumption, it is unclear who is benefiting from a particular pattern of spending. However, Dunbar, Lewbel, & Pendakur (2013) show that allowing for an assumption that preferences for a private assignable good are similar across people allows resource shares to be recovered from a household demand system. I identify resource shares by implementing a linear version of this demand system derived by Lechene, Pendakur & Wolf (2022). As opposed to the original nonlinear formulation, this linear version allows me to use shares identified from Mexico's national household expenditure survey (ENIGH) as an outcome in a staggered difference-in-differences strategy (Callaway & Sant'Anna, 2021) to estimate treatment effects in a two-stage approach.

The testable predictions for the effect of NFUD on resource shares are derived from the limited-commitment

intertemporal collective model (LIC) described in Chiappori & Mazzocco (2017). This model posits that a change to a spouse's outside option can trigger a threat to leave a marriage which can be resolved through commitments to changes in the relationship. The spouse who delivers a credible threat to exit the marriage gains leverage in this process. The model provides two testable predictions for the effect of a policy decreasing the cost of divorce: (a) marriages where a spouse is closest to invoking the threat to leave will be most affected, and (b) the spouse who threatens to leave will gain power to improve their situation within the marriage.

The adoption of NFUD in Mexico in the period from 2008-2020 provides a useful context for testing these predictions. This is because the reform added an additional no-fault mechanism on top of an existing at-fault unilateral divorce (AFUD) option. Whereas a shift from a regime that only allows for mutual consent divorce to one that allows for unilateral options has been extensively modeled following the Becker-Coase Theorem (Becker 1973, Chiappori et al., 2015), a shift from AFUD to NFUD can be summarized as a decrease in the cost of divorce that can be modeled using the LIC (see Brassiola, 2016). Additionally, the pre-treatment availability of AFUD provides a baseline demand for unilateral divorce such that the introduction of NFUD can be modeled as a provision of a cheaper substitute for such demand.

Much of the existing work frames the effect of divorce reform in gendered terms: whether the reform benefited women or men as a group (e.g., Roff, 2017). While this gendered approach is a useful framework, it may mask heterogeneous effects that are just as significant to understand. For example, Fernandez & Wong (2017) find that while women on average benefit from a mutual-consent divorce regime, this is split significantly by income group. Applying the LIC framework to resource shares, it is not immediately clear what the effect of NFUD will be. If the gender of the spouse whose outside option increases in value is the same across households, then the average effect on resource shares may vary in magnitude but not in its direction. The partner who gains leverage due to the reform will increase their share. However, if there is substantial heterogeneity across households as to which partner sees a gain to their outside option, then it is likely that each spouse will strive to increase their own share. This outcome will be missed when averaging the two patterns together.

I implement this approach in three main steps. First, I use administrative divorce records to calculate pre-treatment at-fault divorce filing probabilities. Assuming that no-fault divorce serves as a substitute for

at-fault filings, I test whether the reform's effects vary across couples based on which spouse's marriage participation constraint is binding and to what extent it does so. Second, I separately recover the resource share of children alongside those of the two adults. While bargaining models typically include two Pareto weights that determine each member's bargaining power, resource shares identify expenditures for all household members regardless of whether they have any independent decision-making power. Third, I use a linear approach to estimating resource shares that isolates the slope of the Engel curve as the fraction of marginal household expenditure on each member. This reframing allows for a separable two-stage approach that is, in practicality and interpretability, more compatible with identification of treatment effects with staggered adoption using difference-in-differences.

I find that while the reform on average decreased resource shares of women with children by 4.4 points, it also appears to have decreased men's shares by 3.8 as spending was redistributed to children. Once the children's share is observed directly, the apparent decrease in wives' shares is consistent with the broader evidence that divorce reform strengthens the position of women within the household, given documented gender differences in preferences over spending on children. When disaggregating couples based whether they have minor children, I find that the reform on average had no discernible effect on shares for couples without children.

After estimating the effect of the reform across all households, I test the predictions of the LIC model by replacing the treatment indicator in the resource share estimation step with an interaction between treatment and membership in one of four groups. The groups are defined based on (a) whether a couple's probability of filing for divorce is above/below the median and (b) whether the wife's probability of filing for divorce relative to her husband is above/below the median for the whole sample. Once the shares are estimated, effect sizes come from calculating Callaway & Sant'Anna (2021) group-time ATTs independently for each group.

The results can be summarized across four main findings. First, only groups with children are affected by the reform. Second, among couples with children the previously observed pattern of spending being redistributed from adults to children is maintained. Third, among couples with children the predicted filer increases their resource share relative to their spouse. Fourth, this increase is larger among couples in the higher probability of filing groups compared to those in the lower.

A close paper in this literature is Hoehn-Velasco & Penglase (2021b) which finds that the adoption of NFUD in Mexico did not affect women’s labor force participation. The authors argue that this result, at odds with the broader literature, can be explained by gendered social expectations and labor market conditions that prevented the reform from having its typical effect of improving outcomes for women. To bolster this claim, the paper estimates intrahousehold resource shares and finds that the adoption of NFUD resulted in a decrease in shares for married women. My findings offer a different interpretation that suggests that the effect of NFUD is seen through its effect on children’s resource shares as opposed to adult shares.

My paper makes three main contributions. First, it documents that the adoption of NFUD in Mexico reallocated household resources substantially toward children, with both adult shares declining and the children’s share absorbing the difference. This finding depends on separately recovering the children’s resource share alongside the wife’s and husband’s. Second, it tests predictions of the limited-commitment intertemporal collective model using administrative divorce records to construct measures of each spouse’s outside option. It shows that while effects on adult shares track which spouse held the larger relative gain in their outside option, couples shift bargaining power regardless of how likely the exit threat itself is to be used. Third, this paper joins a small literature that estimates effects of policy on resource shares through linear Engel curves recovered via three-stage least squares, constructing each member’s share as a ratio of slope coefficients rather than through nonlinear GMM. This approach separates resource share recovery from the second-stage treatment-effect estimator, making it directly compatible with the Callaway & Sant’Anna (2021) group-time estimator used for difference-in-differences estimation in staggered treatment contexts.

2 No Fault Unilateral Divorce in Mexico

While divorce laws that allow a single spouse to file for divorce without needing to prove cause (i.e., no-fault unilateral divorce, or NFUD) were enacted in the United States throughout the 1970’s and 1980’s, in Mexico this shift occurred in the 2010’s. In both cases the adoption of NFUD led to a spike in divorce filings with the total number of divorces filed annually doubling in Mexico between 2007-2018, though in the US the trend reversed itself within a decade (Wolfers, 2006).

Reducing restrictions on divorce however affects other outcomes besides divorce rates. Studies focusing

on the US have found that the adoption of NFUD reduced female suicide and domestic violence (Stevenson & Wolfers, 2006), lowered investment in marriage-specific capital (Wolfers, 2007), increased women's labor force participation (Gray, 1998), and incentivized an increase in household savings (Voena, 2015). Many of these same relationships have been investigated by researchers examining the effects of Mexico's divorce reform, though findings have been more mixed regarding domestic violence (Hoehn-Velasco & Silverio-Murillo, 2020; Garcia-Ramos, 2021) and female labor supply (Hoehn-Velasco & Penglase, 2021b). This study most closely follows Hoehn-Velasco & Penglase (2021b) who show that no-fault divorce laws in Mexico marginally increased working hours for employed married women but had no effect on the entire sample of married women. The authors suggest that this mixed result may be driven by more patriarchal social norms and economic structures in Mexico.

One significant difference between the adoption of NFUD in the US and Mexico is at-fault unilateral divorce (AFUD) regime that preceded it. While state laws varied, the barrier for providing a legitimate cause for divorce in the US was relatively high. The practice in New York state of hiring a sex worker together with a photographer to provide evidence of infidelity as cause illustrates the lengths to which dissatisfied spouses had to go.

In Mexico however prior to the reform, spouses could obtain divorce under mutual consent or unilaterally by going through the judicial system and providing one of a set of possible causes for divorce, depending on their state's civil law code (domestic violence, infidelity, and extended separation being some of the most common). In states where extended separation (typically between 6 months and two years) was an allowable reason for divorce, dissatisfied spouses seeking unilateral divorce had a path where they could obtain a divorce unilaterally by leaving their marital home (while still married) and filing for divorce at the end of the waiting period. However, during this waiting period they would not have access to any of the relief provided through divorce law including potential alimony, custody agreements and child support, division of marital assets, or the ability to remarry. By providing for a faster way to dissolve a marriage unilaterally, NFUD allowed spousal resources to be reallocated sooner, benefiting the spouse with fewer resources and allowing both to remarry. Modeling this reform as a change in utility rather than the bargaining framework itself, provides a more realistic portrait of the available tradeoffs.

Temporary separation without divorce can also be a type of internal threat where dissatisfied partners can

use the threat of reversion to a noncooperative equilibrium within the marriage for bargaining (Lundberg & Pollak, 1993). While a non-cooperative separate spheres approach still assumes the marriage remains intact, a further non-cooperative threat within marriage may include one partner packing up and moving out for a period of time.

2.1 Treatment Identification

While Mexico City became the first federal entity to allow for NFUD in 2008, it took several more years until Hidalgo became the first state to follow suit in 2011. In the following years, a number of states resisted the movement to allow for NFUD and some attempted to modify their civil code to explicitly require an approved cause for unilateral divorce. The Mexico Supreme Court delivered a ruling against such laws in 2015, causing a wave of states revising their laws to come into compliance. By 2017, a majority of states had adopted NFUD (Hoehn-Velasco & Penglase, 2021a).

Determining exactly when NFUD is adopted in each state presents challenges due to the nuanced intersection between the legislative, judicial, and administrative branches of government when it comes to family law in Mexico. Researchers reviewing changes to the legal code have disagreed on some dates of adoption (e.g., Aguirre 2019 vs. Hoehn-Velasco & Penglase, 2021b). Additionally, researchers have noted a discrepancy between the time NFUD is legally adopted in the state and a defacto adoption date. NFUD divorces are noted to have occurred in a state before the civil/family code was amended in the state to allow for it. There are also times when the code is amended but NFUD divorces are not observed.

I define a state as treated beginning in the first year in which no-fault unilateral divorce was either (i) legally available (de jure adoption) or (ii) the most common form of unilateral divorce filing (de facto). This takes into account the decentralized administration of family law in Mexico where in some states courts began granting no-fault unilateral divorces before the reform was formally codified while in others legal adoption occurred before NFUD came into wide usage. For most states the two dates coincide or differ by a single year. Taking the minimum ensures that households are not classified as untreated during years in which the no-fault option was already effectively available to them.

The outcome data are repeated cross-sections from a biennial household expenditure survey, so when mapping treatment timing into survey time each state's adoption year is assigned to the first survey wave

conducted with the reform in force. There are three main features of the resulting NFUD adoption schedule. First, adoption is heavily staggered across five cohorts. Second, every state is identified as adopting within the sample window, so there is no never-treated group. The states in the final cohort (Baja California and Tabasco) therefore always serve as comparison units, but not treatment effect can be estimated for them since there is no control group once this cohort has been treated. Third, Mexico City is treated from the first sample period and consequently has no pre-treatment observation. As a result a treatment effect cannot be estimated for households in Mexico city and these households are also excluded from resource share estimation.

Table 1: Treatment Timing by State

State	Jure	Facto	Period	State	Jure	Facto	Period
Mexico City	2009	2008	1	Nayarit	2016	2015	5
Hidalgo	2012	2011	3	Nuevo León	2016	2017	5
Guerrero	2013	2012	3	Puebla	2017	2016	5
Mexico	2012	2012	3	Querétaro	2015	2016	5
Coahuila	2013	2013	4	Sonora	2015	2017	5
Chihuahua	—	2014	4	Tamaulipas	2016	2015	5
Quintana Roo	2018	2013	4	Tlaxcala	2017	2016	5
Sinaloa	2013	2013	4	Baja Calif. Sur	2018	2018	6
Yucatán	2014	2013	4	Chiapas	2017	2019	6
Aguascalientes	2015	2015	5	Durango	2018	2022	6
Campeche	2016	2022	5	Jalisco	2018	—	6
Colima	2017	2016	5	Oaxaca	2018	2017	6
Guanajuato	2016	—	5	San Luis Potosí	2018	2017	6
Michoacán	2017	2015	5	Veracruz	2017	2020	6
Morelos	2017	2016	5	Zacatecas	2018	2017	6
				Baja California	2019	2025	7
				Tabasco	2020	2023	7

Notes: Treatment is defined as the earlier of the de jure (“Jure”) and de facto (“Facto”) year, where the de facto year is the first in which no-fault unilateral divorce becomes the most common form of non-consensual divorce. Entries marked “—” indicate that the corresponding margin was never reached; for these states timing is determined by the other margin alone. The survey period maps the treatment year onto the biennial ENIGH panel: period 1 = 2008, 2 = 2010, 3 = 2012, 4 = 2014, 5 = 2016, 6 = 2018, 7 = 2020, where each period is the nearest survey wave on or after the treatment year. No state adopts in period 2, and Mexico City is always-treated over the 2008–2020 window. Cohort sizes are $n = 1, 3, 5, 13, 8, 2$ for periods 1, 3, 4, 5, 6, 7 respectively.

3 Household Bargaining Model

The structure of the household bargaining model and its interaction with the adoption of NFUD is key to correctly evaluating the possible effects of the reform. Below I outline a theoretical model and discuss how

it is affected by the adoption of NFUD.

3.1 Limited-commitment Intertemporal Collective Model (LIC)

In a collective household model, the members of a household make consumption decisions that are Pareto efficient such that no member can be made better off without another being worse off. However, these decisions are not always made equally by both partners. Individual and social factors affect each partner's bargaining power over consumption decisions. Pareto weights model this process by implicitly indexing the relative significance placed on each partner's preferences (i.e., utility function) in determining the outcome of the household's decision. These weights are typically standardized to sum to one across household members: a Pareto weight of one signifies that the member functions as a dictator in the process while a Pareto weight of $1/N$, where N equals the number of household members, signifies that the preferences of each member are taken into account equally (see Figure 1).

In the short run, the household allocates resources M across private goods $[q_{i,1}, \dots, q_{i,n}] \in Q_i$ and public goods $[g_1, \dots, g_n] \in G$ to solve the optimization problem:

$$\begin{aligned} \max_{Q_1, Q_2, G} \quad & \mu u_1^m(Q_1, G, u_2^m) + (1 - \mu) u_2^m(Q_2, G, u_1^m) \\ \text{subject to} \quad & c(G) + \sum_{n=1}^N \sum_{i=1}^2 p_n q_{i,n} \leq M \quad \& \quad u_i^m \geq u_i^d(X_i) \end{aligned}$$

where p_n is the price of good n that is consumed in the marriage, q_n is the quantity of the good, and Q_i is a vector of quantities of all goods consumed by partner i . The Pareto shares for each partner are indexed by μ_i while u_i^m denotes each spouse's current utility from their marriage. The Pareto shares are a parameter on the optimization problem that determines Q_i , but do not themselves affect each partner's individual utility u_i^m once Q_i and G are determined. Finally, u_i^d denotes each spouse's utility outside of the marriage given their consumption of goods X_i . In the short run, the household can only decide how their consumption across goods is distributed (Q_i). In the long run, bargaining allows households to adjust shares μ_i before solving the maximization problem as part of a limited-commitment intertemporal collective model (Chiappori &

Mazzocco, 2017).

An adjustment of these Pareto weights results in a new solution for the optimization problem. Such an adjustment is required when there are changes in the utility in marriage or the utility in divorce such that there is no solution to the optimization problem that satisfies $u_i^m \geq u_i^d$ for all i . This can be due to changes in preferences over time that affect utility functions $u_i^m()$ and $u_i^d()$, or they can be changes to actual consumption in either state. In this model, the adoption of no-fault unilateral divorce affects consumption of goods X_i through a shock (ω) to one or both partners.

Formally, after solving the first-period optimization problem, the household receives a shock to partner one's X_i in the second period in the form of $\omega \in (\omega_l, \omega_m, \omega_h)$ where

$$\begin{aligned} u_1^m &> u_1^d(X_1\omega_l) \quad \text{and} \quad u_2^m > u_2^d(X_2) \\ u_1^m &< u_1^d(X_1\omega_m) \quad \text{and} \quad u_2^m - (u_1^m - u_1^d(X_1\omega_m)) > u_2^d(X_2) \\ u_1^m + u_2^m &< u_1^d(X_1\omega_h) + u_2^d(X_2) \end{aligned}$$

A shock can occur to both partners and can differ in size, but for simplicity at this time I focus on a shock that only affects partner one. In the first case, the low shock results in partner one continuing to prefer marriage over divorce. In the second medium case, the shock causes partner one to prefer divorce. However, the second partner has a surplus of marital utility (defined as $u_2^m - d_2^d$) that may be transferred to the dissatisfied partner to keep them committed to the marriage. In the third case, the sum of utilities outside of marriage exceed the sum of those in marriage. Even if the second partner prefers to remain married, there is not enough utility that can be transferred to the first partner. Utility is transferred as a result of moving a share of partner two's Pareto weight to partner one, resulting in a solution to the household optimization problem that increases partner one's utility in marriage.

The LIC contrasts with other intertemporal collective household models that feature full commitment and those with no-commitment. In a full commitment model, the Pareto weights are set at some point at time early in the formation of the union, and do not change no matter how the outside option changes. Couples can get divorced if their outside option improves, but they cannot bargain to increase their Pareto weight

(Fernandez & Wong). Conversely, in a model without commitment households bargain over Pareto weights in every period regardless of whether a threat point has been reached.

3.2 LIC in the Case of No Fault Unilateral Divorce

The framework for modeling the adoption of NFUD has typically been to focus on the movement from a regime where consent is required from both partners to one where one partner can unilaterally exit. Such a framework (Becker-Coase Theorem) incorporates commitments to future transfers between the partners in order to buy out an unwilling spouse to obtain a divorce. Crucially, this framework makes no room for bargaining within the marriage because there is no power gained from the potential threat of divorce; someone seeking divorce is either able to or unable to offer their partner enough resources to obtain their consent.

In modeling the adoption of no-fault unilateral divorce as a shock to X_i , I argue that this reform affected the cost of divorce rather than the bargaining structure itself within the marriage. While additional adjustments to the Becker-Coase theorem that offer opportunities for bargaining within marriage have been proposed (Chiappori, Iyigun, & Weiss, 2015), viewing the reform as a change in the cost of divorce is more relevant and realistic in the context of the reform in Mexico.

To illustrate the effect of a change in the cost of divorce, consider the figures below. In figure 1, a couple faces two different utility frontiers when solving the collective household optimization problem based on their combined individual utilities: a marriage frontier (blue) and a divorce (red) frontier. The marked location on the marriage frontier (M) represents the current distribution of utilities between the spouses, while the marked location on the divorce frontier (D) signifies the distribution that would occur after exiting the marriage. The marriage frontier includes all outcomes that are Pareto optimal, those where one spouse cannot be made better off without another being made worse off, a necessary condition for the collective optimization problem. The divorce frontier includes all potential distributions of the partners' utilities outside of marriage that might be mutually committed to or otherwise required by law. In the illustration above, the slope of both curves is -1, meaning that utility is transferred on a one-to-one basis. This is not necessary for the model to hold. It may be that at certain points along either curves that a one unit increase in the wife's utility corresponds to a less than one unit decrease in the husband's utility. It may also be that each partner's utility is complementary, bending the curves into a more concave shape. At this point, I assume that the two curves do not cross.

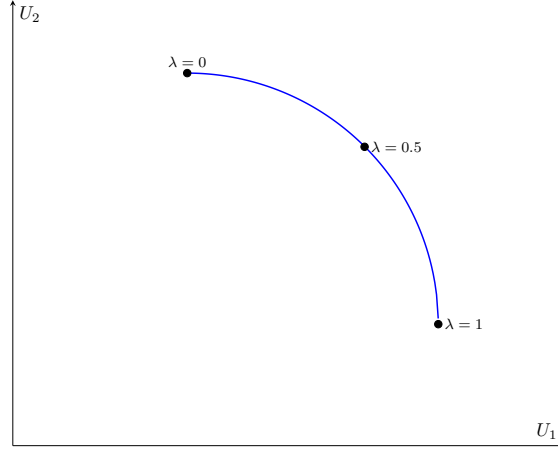


Figure 1: Pareto frontier in marriage with symmetric altruism.
 $\max \lambda U_i + (1 - \lambda) U_j$
 $U_i = \ln(c_i) + \phi_i \ln(c_j) + \psi_i \ln(g)$
 $f(\lambda, \phi, \psi) = c_i, c_j, g$

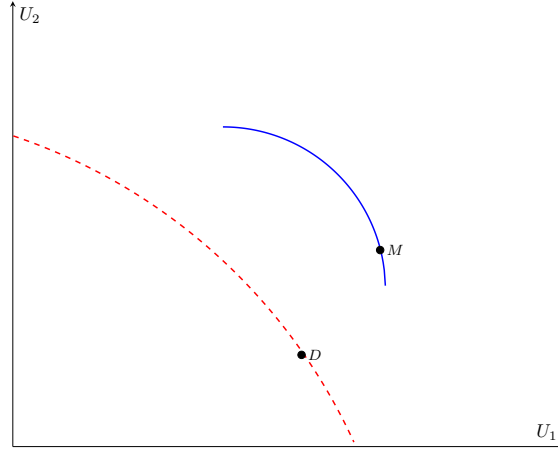


Figure 2: Adding Pareto frontier in divorce. $U_i = \ln(c_i)$
M and D depict a possible set of allocations the couple faces in each state. In this unequal marriage, both partners prefer the marriage outcome. There is no value of D that would result in divorce.

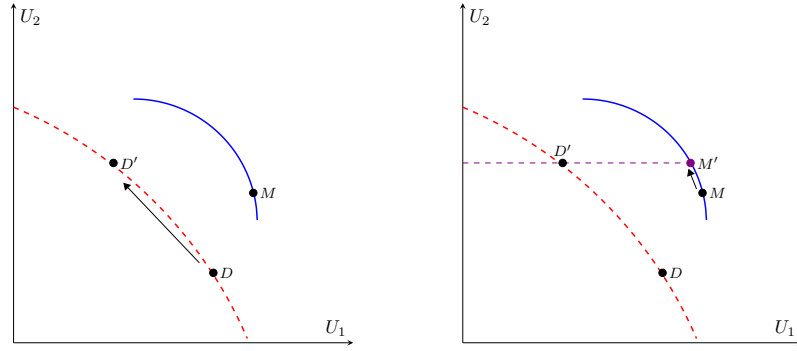


Figure 1: Movement along the divorce frontier results in renegotiation since $U_2(D') > U_2(M)$. Examples: Alimony, division of assets, child custody agreements, and faster/more predictable access to these. To increase partner 2's utility in marriage to equal their utility in divorce, λ is transferred to increase their bargaining power, resulting in M moving to M' . If mutual consent required, there is no opportunity for renegotiation at any value along the D curve.

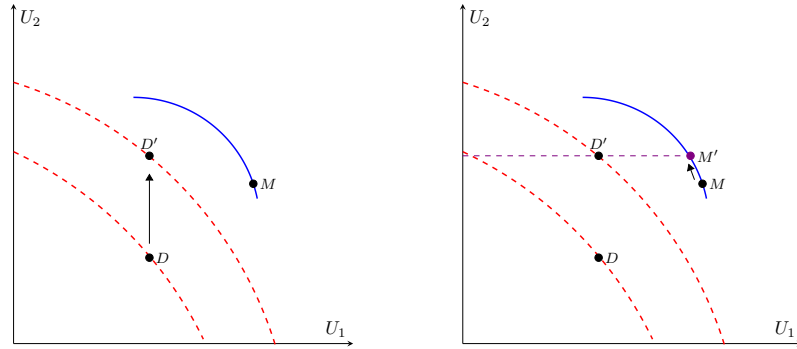


Figure 2: Shift in divorce utility frontier benefiting partner 2 that results in renegotiation. Examples: Simplified process, reduced stigma of divorce, and remarriage opportunities. If mutual consent is required, there is no renegotiation at any value along either D curves (there is no renegotiation if divorce occurs).

The scenario shows the case where the second-period shock causes the divorce distribution (D) to move to the right to D^* . This might represent a more equitable division of assets that a spouse gains access to when filing for no-fault unilateral divorce as opposed to separating and waiting the mandatory 2-year period. In this case, the wife's utility outside of the marriage is now greater than it was inside the marriage. As a result, the marriage utility frontier must move to the right as well until the wife obtains at least the same utility as she now obtains under the new divorce law. A wife who obtains a lot of utility from her marriage must do so because she would receive a lot of utility outside of marriage as well. If not, she does not have the position to retain this amount of utility without transferring it to her husband.

Pareto weights are determined by a number of factors, but are ultimately enforced by the (perceived) threat of a partner refusing to continue to cooperate in the collective household and creating their own separate noncooperative household sphere or leaving the household entirely (i.e., exercising their outside option). Legislation governing divorce, such as increasing alimony payments, can affect Pareto weights by making the collective household decision no longer be Pareto optimal. This increased payoff from divorce would then result in one of the partners being able to make themselves better off by exercising their outside option. In response, the partner who wants to preserve the marriage can allow their partner to have greater say over certain household decisions, redistributing their bargaining power to them, which results in a new Pareto optimal outcome.

Even if it does not cause reallocation at the time of the reform, the new rules governing partners' welfare post-divorce continue to be enforced throughout the length of the marriage and can prompt reallocation at later points in time. It should be noted that the threat to leave does not need to be invoked to carry weight; as long as each partner updates their beliefs about their spouse's welfare if they were to leave the marriage under the new law, they will be motivated to act to avoid the possibility of divorce being invoked in the first place.

3.2.1 Spending on Children

The literature is conflicted on how divorce laws affect household investment in children. Generally, literature focusing on the US argues that making divorce easier, especially when combined with joint-custody laws, reduces overall investment in children (Gruber 2004, Nunley & Seals 2011, Roff 2017). However, the liter-

ature on developing countries paints a different perspective showing an increase in investment (Heggeness 2020, Bargain et al. 2022, Corradini & Buccione 2023) with reductions on barriers to divorce.

One driver of these divergent results may be the role of female labor market participation in shaping what an improved outside option does to child investment. Where female labor force participation is high, a wife's outside option rests in large part on her own market earnings. A stronger bargaining position draws her time toward market work and away from childcare. Where female labor force participation is low, her outside option rests less on employment and more on post-divorce transfers and her position as the primary parent. This can also occur with high female labor force participation if there is equal division of property post-divorce since in this case her outside option also rests on her portion of the divided property (Voena, 2015).

3.3 LIC and Treatment Heterogeneity

The adoption of NFUD affects the post-divorce outcomes for all married couples, but it is unlikely to affect bargaining over resource shares for the majority of marriages. Partners in unions that were very stable before the reform have no reason to renegotiate shares after the reform is adopted since neither partner could credibly threaten to divorce. The reform instead affects marriages where a partner is, prior to the reform, already somewhat indifferent between remaining married or getting divorced. The only thing keeping them married might be that the divorce process is so time consuming, stressful, complicated, and expensive. The simplification in the procedure that the reform relieves these pressures and provides them with a realistic opportunity to divorce. While many will do exactly this, as shown by the large increase in divorces following reform, this might also offer an opportunity for some couples to renegotiate resource shares in a way that satisfies both partners and maintains the marriage.

While it is not possible to observe how much each partner values marriage and divorce in order to identify propensity for treatment, one signal that can be used are contested divorce filings. When one partner files for divorce, they communicate that they prefer the divorce outcome to their current marriage. It also shows that their partner prefers maintaining the current marriage over their divorce outcome. Consider a partner who, after the reform is adopted, determines that the divorce outcome is now preferable to maintaining their

current marriage. There may be room for renegotiation within their marriage however that will increase their resource share within the marriage.

Rather than testing whether unilateral divorce reform increases the bargaining power of a particular member of the household (husbands or wives) as a group, I predict that the reform benefits partners who have the greatest increase in the value of their outside option due to the reform since they are the ones who will be able to credibly threaten to leave the marriage. While some of the benefits that come with no-fault unilateral divorce when compared with fault-based divorce or informal separation – such as alimony, child support, child custody, and division of marital assets – are typically gendered in distributing resources post-divorce from men to women, other benefits such as reduced uncertainty about the outcome, less time invested in the legal process, as well as the ability to remarry (and possible long-term shifts in acceptance of divorce) can benefit spouses regardless of gender. This means that it is crucial to identify which partner obtains the power to credibly threaten to leave in each couple. Otherwise, the change will be biased towards zero since any increase in women's bargaining power within some couples will be counterbalanced by an increase in men's bargaining power in others.

While standard approaches to unilateral divorce reform model the change as a shift from requiring mutual consent for divorce to making it a single-person decision, this is not a fully accurate description of the change in Mexico since it was possible for a dissatisfied spouse not only to exercise her outside option by simply leaving, but also obtain a unilateral divorce after a period of separation. Instead, the reform accelerates the process of divorce, which in turn speeds up the process of obtaining relief such as alimony, child support, and division of marital assets. In the former model, divorce under mutual consent is not attainable for a spouse who cannot pay-off their partner to acquiesce to the divorce (Becker-Coase Theorem), but in the latter it is if the divorce-seeking partner can forego any institutional payoffs from divorce until the end of the required waiting period. They will therefore only choose to pay-off their partner and obtain a mutually consented divorce if the cost involved in waiting is greater than the required pay-off amount. Knowing this, their partner will demand a higher payout if the cost of waiting is large, but can only demand a small amount if the cost of waiting is low (or no amount if the cost of waiting is negligible). The reform may therefore particularly benefit women with low earning potential since they have the greatest cost in waiting to obtain

a divorce. Additionally, a spouse looking to get remarried quickly may also benefit from a speedier process.

The reduction in the burden of proving fault affects X_i through two main channels: faster and more affordable access to legal protections afforded under divorce and changes to the social cost of divorce. Some of these may affect the utility distribution between the partners (e.g., alimony, asset division) as shown in Figures 3(a) and 3(b). Others may increase the total utility in divorce for one or both partners (e.g., simplified divorce process, less uncertainty, better opportunities for remarriage) as depicted in Figures 4(a) and 4(b). These new divorce utilities result in a wide range of outcomes depending on the particular marriage. Some couples will immediately divorce, while others may be pushed away from the brink of divorce (such as when divorce becomes a worse bargain for the partner contemplating it). Some couples will immediately have to reallocate Pareto shares to maintain their marriage, while others will be affected only further down the line in their marriage.

Given these expected effects are highly heterogeneous by household, much of the research has focused on the effect the reform has had on specific groups. Hoehn-Velasco & Silverio-Murillo (2019) find that the reform had no change on female suicide or homicide rates, in contrast to findings by Stevenson & Wolfers (2006). Moreover, they corroborate conclusions from related studies (Silverio-Murillo, 2019; Garcia-Ramos, 2019) which show that intimate partner violence between married couples may have increased following the liberalization of divorce laws. In Hoehn-Velasco & Penglase (2021a), the authors show that women became significantly more likely to file for divorce under the unilateral no-fault regime. However, in Hoehn-Velasco & Penglase (2021b), the reform is found to have no effect on married women's labor force participation, but associated with a small increase in hours worked by employed married women. Contrasting these results to those in Stevenson (2008), the authors hypothesize that gendered cultural expectations and labor markets modulate the effect of the reform on women's bargaining power.

Given the marriage participation constraint, the largest gains in marital utility (Δu_i^m) are among those where the change in utility in divorce (Δu_i^d) is greatest over their pre-reform marital surplus ($u_i^m - u_i^d$).

Formally:

$$\begin{aligned}
\Delta u_i^m &= \Delta u_i^d - (u_i^m - u_i^d) \rightarrow \\
&\quad (u_i^{dR} - u_i^d) - u_i^m + u_i^d \rightarrow \\
&\quad u_i^{dR} - u_i^m \\
\text{subject to: } &u_i^m \geq u_i^d \quad \forall i \in i, j \\
&u_j^m - (u_i^m - u_i^{dR}) \geq u_j^{dR}
\end{aligned}$$

Where u_i^{dR} identifies the post-reform utility in divorce.

The largest increase in marital utility would therefore be in cases where Δu_i^d is very large and where the spouse has very little marital surplus to begin with such that $u_i^m - u_i^d = 0$. Here, $\Delta u_i^m = \Delta u_i^d$. In other words, these are people who were already on the threshold of initiating the divorce process, where the reform further significantly increased their utility in divorce such that they could now afford to leave, and where their spouse had enough marital surplus that they were willing to part with in order to maintain the marriage.

Given the number of factors that each individual would consider, many of which cannot be unobserved, I instead proxy for this change in marital utility by using data from divorced couples. The goal is to train a model to estimate the likelihood of being treated by the reform. While a simple approach would be to use all of the divorce data to train the model on who files for unilateral divorce, it's important to avoid contaminating the DiD identification strategy with composition shifts. Since the reform will change who files for no-fault unilateral divorce, as well as who initiates, including post-reform data in the training model will cause the score to be partly an outcome of treatment. For example, if states where a higher proportion of the population has a college education are more likely to get the reform earlier, then the training data may incorrectly show that higher education predicts higher no-fault divorce rates.

One option is therefore to train the model only on pre-reform data. This is possible using the de facto treatment definition, but is not feasible with the de jure treatment. Another option is to train the model on all divorces, but include a treatment indicator. Then, generate a pre-reform and post-reform score for the married couple sample and take its difference. The third option, which can be used with the de jure treatment definition, is to consider at-fault unilateral divorces as close substitutes to no-fault.

3.4 Data Sources and Estimation Procedure

Three data sources are used for estimating the probability of AFUD filings and the identity of the filer: a 2010 Mexico census person-level sample obtained from IPUMS-International (census data), annual divorce-level filing data from the INEGI's online statistical database (divorce data), and municipal-level statistics from the Sistema Nacional de Información Municipal (covariate data).

The divorce data consists of about 600,000 cases that are used to estimate filing probabilities. This includes divorces that were either based on mutual consent or filed under one of three separation-based AFUD types. Additionally, the divorce must have occurred in a state before NFUD was adopted and not be missed any of the predictor variables from the model (e.g., the ages of divorcees and the number of minor children).

I use these sources to generate three estimates for ENIGH sample households: (a) the probability of a married couple getting divorced, (b) the probability if a couple gets divorced that it is under one of three separation-based AFUD types, and (c) the probability if the divorce is one of these AFUD types that it is the husband/wife who files it.

To obtain the probability for (a) I restrict the census data to persons who are married or divorced. I then append to it the ENIGH sample of married couples and add municipal covariate data. I run a linear prediction model for divorced status on the census sample (separately for men and women) and generate predicted divorced status on the ENIGH sample.

To obtain the probability for (b) I restrict the divorce data to divorces filed under either mutual consent or one of the three separation-based AFUD reasons in a state prior to the adoption of NFUD. Just as it was done for (a) I append the ENIGH sample of married couples and add municipal covariate data. I run a linear prediction model for AFUD filing on the divorce sample and generate predicted AFUD filing on the ENIGH

sample.

To obtain the probability for (c) I restrict the divorce data to divorces filed under one of the three separation-based AFUD reasons. I then identify which partner filed for divorce. Following the same steps as in (a) and (b) I run a linear prediction model separately male and female filers and generate for each ENIGH household the probability of (c) for the wife and for the husband.

To obtain the probability of AFUD occurring in a married ENIGH household, I evaluate the maximum of the product of (a) and (b) for each partner in the household. If NFUD functions as a lower cost substitute for AFUD then this should predict the likelihood that the household is affected by the reform. To obtain the relative likelihood of each partner filing for divorce I take the value of (c) for the wife and normalize it by dividing it by the sum of (c) for both partners. This should predict which partner is more likely to gain bargaining power following reform. I split households based on whether they lie above or below the median for each of these measures, resulting in four divorce groups: (1) Low Probability + Husband Likelier, (2) Low Probability + Wife Likelier, (3) High Probability + Husband Likelier, and (4) High Probability + Wife Likelier

4 Estimating Resource Shares

Pareto weights (μ_i) are not observed and while decision-making power may be interpreted through direct interviews with household members, a more comparative measure tied to individual welfare is more suited for large-scale study. Resource shares (η_i), defined as the share of total household consumption by individual i are a common avenue of approach. A household where the majority of consumption goes to a single member is one where decision-making power is less equal than a similar household where consumption is split more evenly. Some goods are expected to be consumed unequally based on the preferences and needs of each member, but goods that are desired and required somewhat equally are expected to be consumed somewhat equally unless a member with significantly more bargaining power intervenes.

Table 2: Prediction Model (a): Probability of Any Divorce (Census)

Individual controls	Men	Women
Children: 1 minor child		−0.0101***
2+ minor children		−0.0315***
Education: Primary	−0.0012	−0.0091**
Lower secondary	0.0022	0.0060
Upper secondary	0.0102**	0.0218***
Some college	0.0106	0.0356***
College	0.0005	0.0102
Postgraduate	0.0036	0.0464***
Municipal controls		
Urban type: Mixed	−0.0322***	−0.0265**
Rural	−0.0237***	−0.0173*
Semi-urban	−0.0294***	−0.0188*
Large Urban	−0.0304***	−0.0205**
Medium Urban	−0.0329***	−0.0189**
High School Completion (Overall)	0.0269***	0.0237***
High School Completion (Women)	−0.0231***	
Employment Rate (Overall)	−0.0400*	−0.1446***
Employment Rate (Women)	0.1154***	0.2686***
State controls		
Separation days	−0.000039**	0.000284***
Constant	0.0268**	−0.0794***
R^2	0.0071	0.0192
Observations		

Notes: Linear probability estimates of an indicator for divorced status on the pooled IPUMS census sample of married and divorced persons, run separately for men and women, and used to predict each ENIGH couple's divorce probability (prediction (a)). **Separation days** is the state-level median of how long (in days) AFUD divorce filers on separation grounds wait between the start of the separation and the finalization of the divorce. **Remarriage Ratio** terms are the number of unmarried persons of the opposite sex within a five-year age range in the municipality as a ratio for each spouse. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 3: Prediction Model (b): Probability of Unilateral (AFUD) Divorce and (c): Probability the Wife is the Filer

	(b) Unilateral filing	(c) Wife files
Individual Controls		
Children: 1 minor child	0.000414	0.0739***
2+ minor children	−0.0587***	0.0436**
Wife’s age	0.0073***	0.0094***
Wife’s age ²	−0.5494***	−1.0692***
Wife’s Remarriage Ratio	−0.0241***	0.0478***
Husband’s age	0.0081***	−0.0049***
Husband’s age ²	−0.5802***	
Husband’s Remarriage Ratio	−0.0252***	0.0159***
Municipal Fixed Effects		
High School Completion (Women)	−0.1424***	−0.1327***
Employment Rate (Overall)	−0.7284***	−0.3072***
Employment Rate (Women)	−0.5713***	0.5389***
High School Completion (Overall)	0.1349***	0.1366***
Urban type: Mixed	−0.0391***	−0.0533***
Rural	−0.0793***	−0.0331**
Semi-urban	−0.0898***	−0.0400***
Urban (large)	−0.0176***	−0.0229***
Urban (medium)	−0.0457***	−0.0243**
State Fixed Effects		
Separation days	−0.0017***	−0.000078
Interaction Terms		
One child × wife’s age	0.000181	−0.0029***
2+ children × wife’s age	−0.0011***	−0.0029***
One child × husband’s age		0.0034***
2+ children × husband’s age		0.0039***
Number of Children × Urban		
Metropolitan	0.0369***	0.0033
Mixed	−0.0144***	0.0056
Rural	−0.0356***	−0.0129*
Semi-urban	−0.0128***	0.0031
Urban (large)	0.0253***	0.0038
Constant	1.1765***	0.4624***
R^2	0.0640	0.0340
Observations	614,713	142,757

Notes: Linear probability estimate of an indicator for a separation-based at-fault unilateral (AFUD) filing on the pre-reform administrative divorce sample (mutual-consent or AFUD cases in states before NFUD adoption), used to predict each ENIGH couple’s AFUD propensity (prediction (b)); this feeds the contested divorce-propensity split. Linear probability estimate of an indicator for the wife being the filer, on the administrative sample of separation-based AFUD divorces, used to predict which spouse files (prediction (c)) and to form the wife-filing split. The corresponding husband-filing model is the exact complement with the opposite sign. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.1 Linear Identification of Resource Shares

Complete data on consumption is often not available, particularly since it requires determining individual consumption of household public goods. However, Dunbar, Lewbel, and Pendakur (2013), hence referred to as DLP, show that resource shares can be estimated using the share of household spending for a private assignable good. Setting aside the role of children for now, consider the simplified case of a household consisting of two individuals of type t : an adult woman (w) and an adult man (m). Each of them only consumes their corresponding type of private assignable good. Then, household expenditure for each private assignable good can be written as:

$$W_{ht} = \eta_{ht} w_{ht} (\eta_{ht} y_h, p_{ht}) \quad \text{where} \quad \eta_{ht} = f(y_h, p_{ht}, d_{ht})$$

where W_{ht} is total expenditure of household h on good t , y_h is total household expenditure, d_{ht} is a distribution factor (things that affect resource shares that are not prices or expenditures), and $w_{ht}()$ is person t in household h 's demand function for good t . In other words, each partner selects expenditure on good t as if they have a budget of y_h times their resource share η_{ht} .

Estimating the demand function would typically require observing relationships between demand and prices. DLP simplify this by rewriting the household expenditure function into Engel curve form where the relationship between y and w_t is for a given price p . To do so, they assume that the resource share function $\eta_{ht}()$ is independent of household expenditures. As a result, the household expenditure function can be simplified to:

$$W_{ht}(y_h) = \eta_{ht} w_{ht}(\eta_{ht} y_h) \quad \text{where} \quad \eta_{ht} = f(p_{ht}, d_{ht})$$

The household demand function W_{ht} can now be interpreted as an Engel curve that depicts how demand for good t changes as a function of total expenditure y keeping the resource shares (and prices) constant. DLP next apply a Similar Across People (SAP) assumption that the shape of w_{ht} includes a component that is common across all people regardless of type t that is linear in the natural log of total expenditure. Formally,

the assumption can be written as:

$$w_{hts}(\eta_{hts}y_h) = \alpha_{ht} + \beta_s \ln(\eta_{hts}y_h)$$

where s indexes a household type, α_{ht} provides the preferences for good t in household h , and β_s is the common preference for good t across all people across all households of type s . Substituting into the household expenditure function and expanding terms, we get:

$$W_{hts} = \eta_{hts}\alpha_{ht} + \eta_{hts}\beta_s \ln(\eta_{hts}) + \eta_{hts}\beta_s \ln(y_h) \quad (1)$$

Equation (1) shows that the observed budget share for assignable good t has a slope component that varies linearly with $\ln(y)$ and an intercept or level shift component that does not. The slope tells us how the household allocates expenditures at the margin, while the level provides variation not due to changes in expenditure. Using this functional form, we see that a change that affects resource shares η_{hts} will necessarily affect both the slope and the level component. However, the SAP assumption allows us to use the slope component to identify an aggregated η_{ts} .

Since β_s is common for all people in households of type s , η_{ts} can be identified by comparing the slopes of the Engel curves of the two members of the household. We start by taking the derivative of W_{hts} w.r.t. $\ln(y)$:

$$\frac{\partial W_{hms}}{\partial \ln y_h} = \eta_{ms}\beta_s \quad \frac{\partial W_{hws}}{\partial \ln y_h} = \eta_{ws}\beta_s \quad (2)$$

Since by definition $\eta_{ms} + \eta_{ws} = 1$, we can isolate β_s :

$$\frac{\partial W_{hms}}{\partial \ln y_h} + \frac{\partial W_{hws}}{\partial \ln y_h} = \eta_{ms}\beta_s + \eta_{ws}\beta_s = \beta_s(\eta_{ms} + \eta_{ws}) = \beta_s \quad (3)$$

Identifying β_s in turn allows us to isolate η_{ts} from the derivative:

$$\frac{\partial W_{hts}}{\partial \ln y_h} * \frac{1}{\beta_s} = \frac{\eta_{ts}\beta_s}{\beta_s} = \eta_{ts} \quad (4)$$

SAP, the invariance of resource shares and Engel slopes to total expenditure, as well as the ability to find a private assignable good are all significant assumptions in this model.

Starting with SAP, the assumption can be restated as requiring the demand for the good that is being modeled to respond in the same way to changes in the household member's budget share $\eta_{hts}y_h$ across all members of households of type s . In a representative household where resource shares are equal, an increase in total expenditure results in both the husband and wife equally increasing their consumption of their private assignable good (though their overall levels may differ). If spending does not increase equally, then by construction the resource shares will be unequal. DLP test SAP by comparing consumption by single men and women; they find that the restrictions hold as the income elasticities of demand for the private goods do not statistically differ.

4.2 Empirical Estimation of Resource Shares

Equation (4) shows that the resource share can be identified by comparing the first derivative of the budget share equation w.r.t. total expenditure for each partner. The derivative of the budget share equation for each partner's good, $\eta_{ts}\beta_s$ can be identified by regressing $\ln(y)$ onto the observed budget share. This can be accomplished using a seemingly unrelated regression (SUR) in each approach. Note that in the regression specifications that follow, I use $\hat{\beta}_t$ to denote the regression coefficient on $\ln(y_h)$, which estimates $\eta_{ts}\beta_s$ from the SAP demand system.

Below, I outline the econometric strategy I use to estimate resource shares. I first outline a standard strategy that does not use a treatment indicator and then describe why adding a treatment indicator is necessary for causal inference.

4.2.1 SUR Estimation of Resource Shares

Suppose that treatment is not included in the estimation of resource shares. We run Seemingly Unrelated Regression with two equations:

$$\text{Budget Share } W_{ht} = \alpha_t + \hat{\beta}_t \ln(y_h) + \gamma_t S_h + \delta_t Z_h + \phi_t(S_h * y_h) + \rho_t(Z_h * y_h) + \epsilon_{ht} \quad (5)$$

$$\text{where } \phi_w = -\phi_m \quad \text{and} \quad \rho_w = -\rho_m \quad (6)$$

where y is total household expenditure, S indexes household types, and Z is a vector of individual preference controls. There are four different household types S based on the number of co-resident children under the age of 18 (0, 1, 2, 3+). I use this to categorize households for the SAP assumption that the slope of the Engel curve is the same for all people in households of type S . Controls Z include the age of each partner, each of their education level, and an indicator for whether the municipality that they reside in is classified as rural or semi-urban (i.e., not urban). There is an additional restriction within the SUR that the coefficients on ϕ and ρ must sum to zero across partners. Since resource shares must sum to one, the marginal effects of any covariate on the Engel curve slopes must net out across partners. If a covariate increases the slope of the husband's Engel curve by some amount, it must decrease the slope of the wife's curve by the same amount, since the slopes together must continue to sum to β_s .

From the regression, we obtain $\hat{\beta}_t$, which is our estimate of $\eta_{ts}\beta_s$ conditioned on Z , S , and interactions between these and total expenditure y . The resource share is then constructed by taking the ratio of the $\hat{\beta}_t$ for each partner evaluated at the specific s and z values:

$$\eta_{ts}(z) = \frac{\hat{\beta}_t + \phi_t(S_h) + \rho_t(Z_h)}{\hat{\beta}_t + \phi_t(S_h) + \rho_t(Z_h) + \hat{\beta}_{-t} + \phi_{-t}(S_h) + \rho_{-t}(Z_h)}$$

Since $\phi_w = -\phi_m$ and $\rho_w = -\rho_m$, the denominator can be simplified to $\hat{\beta}_t + \hat{\beta}_{-t}$:

$$\eta_{ms}(z) = \frac{\hat{\beta}_m + \phi_m(S_h) + \rho_m(Z_h)}{\hat{\beta}_m + \hat{\beta}_w} \quad \text{and} \quad \eta_{ws}(z) = \frac{\hat{\beta}_w + \phi_w(S_h) + \rho_w(Z_h)}{\hat{\beta}_m + \hat{\beta}_w}$$

These resource shares $\eta_{ts}(z)$ are calculated for each of the two partners in each household and are the primary outcome of interest. In the case where these shares are “treatment blind”, variation in resource shares across households comes from the two interaction effects between household size s and covariates z with total expenditure. Households with similar values for s and z are used to fit Engel curves for both the husband and the wife. Note that while household-level total expenditure and budget shares are used to estimate these parameters, they themselves do not enter the resource share calculation. Households with identical s and z values will have identical $\eta_{ts}(z)$ values regardless of their total expenditure and budget shares.

Suppose s indexes the household size mentioned earlier and z consists of the ages of both partners. Then, $\eta_{ts}(z)$ will only vary from the effects of these three values on the slope of the Engel curve. Now if all three of these values are spread randomly between treatment and control groups, resource shares would be constructed by age and household size, averaging out any potential effects of treatment on the share. Further, when comparing these shares using a difference-in-differences approach, any observed effects would be the result of idiosyncratic differences between how s and z are distributed across the treatment and control group. For this reason, resource shares cannot be treatment blind and must be allowed to depend on treatment within the SUR estimation step.

A potential misunderstanding is that recovering η_{ts} requires a fully specified demand model for the assignable good. However, η_{ts} is identified directly from the ratio of assignable-good Engel curve slopes across household members. An Engel curve could be estimated by regressing the clothing budget share on log total expenditure with no additional controls, which would provide resource shares that are constant across all persons of type t in households of type s . However, we would like resource shares to vary across households since we expect this to better reflect differences in bargaining power. To let shares vary, I interact log expenditure with a compact set of demographics (each spouse’s age and age-squared, education, a rural indicator, child-count indicators, and treatment groups) that may affect bargaining power between household members. Separately, I include level-only controls that shift the height of the clothing budget share at a given expenditure, including region and year fixed effects (and the demographics in levels).

Table 4: First-Stage Demand System: Common Treatment Factor

Assignable-good budget share	With children			Without children	
	Wife	Husband	Child	Wife	Husband
$\ln(\text{Total Expenditure})$	0.0028 (0.0018)	0.0037 (0.0019)	0.0075** (0.0027)	0.0080*** (0.0021)	0.0054* (0.0021)
$\ln(\text{Total Expenditure}) \times \text{Treated}$	-0.0006* (0.0003)	-0.0005 (0.0003)	0.0011** (0.0004)	-0.0001 (0.0004)	0.0001 (0.0004)
× Wife's age	0.0000 (0.0001)	-0.0004** (0.0001)	0.0004 (0.0002)	-0.0000 (0.0001)	0.0000 (0.0001)
× Husband's age	0.0001 (0.0001)	0.0004** (0.0001)	-0.0005** (0.0002)	-0.0000 (0.0001)	0.0000 (0.0001)
× Wife's age ²	-0.0052 (0.0152)	0.0526** (0.0166)	-0.0474* (0.0228)	-0.0021 (0.0127)	0.0021 (0.0127)
× Husband's age ²	-0.0127 (0.0128)	-0.0441** (0.0140)	0.0567** (0.0192)	0.0023 (0.0119)	-0.0023 (0.0119)
× Wife's education	0.0004*** (0.0001)	-0.0000 (0.0001)	-0.0004* (0.0002)	0.0002 (0.0002)	-0.0002 (0.0002)
× Husband's education	-0.0002* (0.0001)	0.0004*** (0.0001)	-0.0002 (0.0002)	-0.0002 (0.0001)	0.0002 (0.0001)
× Non-urban	-0.0001 (0.0003)	0.0003 (0.0003)	-0.0002 (0.0005)	-0.0003 (0.0005)	0.0003 (0.0005)
× Two children	-0.0008** (0.0003)	0.0003 (0.0003)	0.0005 (0.0004)		
× Three or more children	-0.0005 (0.0003)	0.0004 (0.0004)	0.0002 (0.0005)		
Level controls	Yes	”	”	”	”
Region FE	Yes	”	”	”	”
Survey-year FE	Yes	”	”	”	”
Budget Share Average	0.0089	0.0085	0.0356	0.0134	0.0124
R^2	0.058	0.036	0.057	0.069	0.048
Observations	42,144	42,144	42,144	16,493	16,493

Notes: Three-stage least-squares estimates of the assignable-good Engel curves for the with-children (three-equation) and childless (two-equation) samples. Each column is the budget-share equation for one member's assignable good; resource shares are recovered as ratios of the estimated $\ln y$ slopes (Lechene, Pendakur & Wolf 2022). $\ln(\text{Total Expenditure}) \times \text{Treated}$ is the constrained treatment distribution factor. Intercept-shifting level controls (demographics, the level treatment term, child-count indicators), region fixed effects, and survey-year fixed effects are included but suppressed. Cross-equation adding-up constraints force the $\ln y$ interaction coefficients to sum to zero across members. Total expenditure is instrumented with income. Analytic 3SLS standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 5: First-Stage Demand System: Treatment by Predicted-Filer Group

Assignable-good budget share	With children			Without children	
	Wife	Husband	Child	Wife	Husband
$\ln(TotalExpenditure)$	0.0028	0.0037	0.0078**	0.0084***	0.0051*
$\ln(TotalExpenditure) \times Treated$					
by Divorce Groups:					
× Low, Husband	−0.0006	−0.0004	0.0010	0.0003	−0.0003
× Low, Wife	−0.0003	−0.0006	0.0009	0.0186	−0.0186
× High, Husband	−0.0010	0.0000	0.0010	−0.0000	0.0000
× High, Wife	−0.0006	−0.0008	0.0013*	−0.0029	0.0029
$\ln(TotalExpenditure)$					
× Wife's age	0.0000	−0.0004**	0.0004	−0.0000	0.0000
× Husband's age	0.0001	0.0004**	−0.0005**	−0.0000	0.0000
× Wife's age ²	−0.0039	0.0523**	−0.0484*	−0.0025	0.0025
× Husband's age ²	−0.0129	−0.0441**	0.0569**	0.0035	−0.0035
× Wife's education	0.0004***	−0.0000	−0.0004*	0.0002	−0.0002
× Husband's education	−0.0002*	0.0004***	−0.0002	−0.0002	0.0002
× Non-urban	−0.0001	0.0003	−0.0002	−0.0003	0.0003
× Two children	−0.0008**	0.0003	0.0006		
× Three or more children	−0.0006	0.0003	0.0003		
Level controls	Yes	”	”	”	”
Region FE	Yes	”	”	”	”
Survey-year FE	Yes	”	”	”	”
Budget Share Average	0.0089	0.0085	0.0356	0.0134	0.0124
R^2	0.058	0.036	0.058	0.069	0.048
Observations	42,144	42,144	42,144	16,493	16,493

Notes: As in Table 4, but the treatment distribution factor is interacted with membership in four predicted-filer groups. Group 1 = Low propensity, Husband filer; Group 2 = Low, Wife; Group 3 = High, Husband; Group 4 = High, Wife. Resource shares are recovered as ratios of the $\ln(TotalExpenditure)$ slopes. Intercept-shifting level controls (including the group-specific level treatment terms and child-count indicators), region fixed effects, and survey-year fixed effects are included but suppressed. Cross-equation adding-up constraints force the $\ln y$ interaction coefficients to sum to zero across members. Total expenditure is instrumented with income. Analytic 3SLS standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.2.2 Instrumenting SUR Estimation

The budget share equations cannot be estimated consistently by SUR because total household expenditure appears on both the right and left-hand side (in the denominator of the budget share) of each equation. Additionally, household expenditure in survey data is measured with error due to recall bias and from the infrequency of purchases. Measurement error in y_h therefore enters both sides and biases the estimated Engel curve slopes.

Following standard practice in the literature, I instrument expenditure with household income in a system of equations in three-stage least squares (3SLS). Reported income is correlated with true household expenditure but is collected independently of it, so recall and infrequency errors in measured spending are plausibly unrelated to it. Additionally, every interaction between a covariate and $\ln(y_h)$ (i.e., the terms $S_h * y_h$ and $Z_h * y_h$) including the treatment interactions inherit the measurement error in expenditure. Each endogenous interaction is therefore instrumented by the corresponding covariate interacted with log income, so that the instrument set mirrors the endogenous set for each term.

4.2.3 Expenditure Data

The ENIGH is a household survey conducted by Mexico's National Institute of Statistics and Geography (INEGI) in every even-numbered year since 1992. The survey collects information about how much money was spent by household members on various categories of goods over the preceding month as well as sociodemographic information. This allows me to obtain the household's clothing budget shares, that is the fraction of household spending for men's, women's and children's clothing, as well as total household expenditure and income, the ages and education levels of household members, as well as their state and municipality of residence. The initial sample includes approximately 230,000 households across six waves of the survey (2008-2018). I restrict my sample to households that include exactly two married adults (i.e., householder and their spouse) and any number of children of the householder under the age of 18. I also drop households in Mexico City since this group is treated for the entire period that I observe. As a result, I am left with about 61,000 total households.

5 Causal Inference Strategy

Since states adopted no-fault unilateral divorce at different times between 2008 and 2020, and because treatment effects may vary both across adoption cohorts and over time since adoption, I estimate group-time average treatment effects rather than a two-way fixed effects regression. This section defines treatment timing, describes the estimator and its identifying assumptions, and details the inference procedure, which must account for the fact that the outcome variable is itself an estimate.

5.1 Identifying Assumptions

Identification requires parallel trends to hold for each treatment cohort. This means that in the absence of the reform, the average resource shares of each cohort would have evolved in parallel with those of the later treated comparison states. The design additionally assumes limited anticipation: household resource allocation does not respond to the reform before it becomes effectively available. The de facto component of the treatment definition works in favor of this assumption, since states where courts moved ahead of statutes are coded as treated when practice changed rather than when the law did. Finally, the comparison across states requires that adoption in one state does not affect allocations in untreated states (SUTVA). Since divorce is governed by the law of the state of residence and the ENIGH samples resident households, cross-state spillovers would have to operate through migration channels. I use the Callaway & Sant’Anna group-time ATT(g, t) estimator since with staggered adoption and effects that vary by cohort and event time, a two-way fixed effects regression assigns negative weights to some group-time cells and incorrectly uses already-treated states as controls for later adopters.

5.2 Group-Time treatment effects with staggered adoption

Standard difference-in-differences effect estimates are identified as the gap between the difference in the treatment and control group immediately prior to treatment and the difference between these groups after treatment occurs. This forms the basis for the group-time ATTs in Callaway & Sant’Anna, but there are additional steps specific to this approach.

First, I set the baseline comparison period to always be the period immediately preceding treatment ($g-1$).

The $ATT(g,t)$ estimate for group g at time t is therefore identified as the gap between the difference in the treatment and control group in $g - 1$ and the difference between these groups in period t .

Second, the control group varies for each $ATT(g,t)$ estimate. Specifically, each unit enters the control groups evaluated at t and $t-1$ if its own treatment date (g') satisfies both of the following conditions: (a) they are treated after the cohort they are being compared to (i.e., those treated in $g' > g$), and (b) have not yet been treated (i.e., $g' > t$). Note that for post-treatment estimates condition (b) binds while for pre-treatment estimates condition (a) binds. This prevents the cohort whose effect is being estimated from entering both the treatment and control groups in pre-treatment estimates.

Using six waves of the survey and five treatment cohorts, I generate a total of 20 group-time treatment effects (there is technically no $ATT(g,1)$ or $ATT(5,t)$). These are then averaged with each $ATT(g,t)$ weighted based on its share of treated observations to produce a "simple" summary effect and an "event-time" effect. The "simple" effect is the average of the post-treatment $ATT(g,t)$ s. The "event-time" effect collects $ATT(g,t)$ s sharing the same event-time, denoted by the difference between g and t . For example, with five treatment cohorts the effect for two periods after treatment is the weighted average of $ATT(2,3)$, $ATT(3,5)$, $ATT(4,6)$, $ATT(5,7)$, and $ATT(6,8)$.

5.3 Bootstrapping with Generated Outcomes

Since resource shares are estimated from the demand system rather than observed, using them as the DiD outcome without additional correction will result in a generated-regressor problem. The first stage applies a common demand system structure, so that every resource share is derived from the same set of estimated coefficients. Sampling error in those coefficients is therefore shared across all observations, which violates the independence assumption underlying the DiD estimation of standard errors. If treatment affects resource shares, this sampling error will affect treated and control groups differently, further increasing the uncertainty of the true effect of treatment. Because DiD compares households in treated and control states, the shared error resulting from the first stage induces within-state correlation in the DiD. While clustering in DiD fixes correlation within that stage, the first stage error remains and affects the entire sample while not being observed through clustering in DiD. I therefore run an end-to-end state-level cluster bootstrap that re-estimates both the demand and DiD stages in each replicate. Each of 999 replicates draws observations from 31 states

with replacement; any repeated draws are labeled as distinct clusters in the DiD step. The predicted-filer group assignments are held fixed at their full-sample cutpoints across replicates.

6 Results and Discussion

6.1 Effects on Resource Shares

Table 6 reports the aggregated group-time average treatment effects on the treated (ATT) across five different sample specifications. Point estimates are from the non-bootstrapped original sample. Households are divided between those with children and those without children and ATTs are reported for each of the groups as a whole as well as for four subgroups for each. The table reports bootstrap standard errors that combine the errors from the resource share estimation stage with the group-time ATT estimator through state-level resampling. The bootstrap SEs are significantly larger than the errors from the group-time ATT estimator on its own, showing that the imprecision in measuring resource shares affects the minimum detectable effect of the reform on shares.

While the treatment effects are significant for a majority of cells under errors from the group-time ATT estimator on its own, only three cells show significant effects under the bootstrap approach (where 95% of bootstrap replicates show a significant effect); these are identified with a star in the table. While these deserve special attention, the imprecisely estimated effects however are still very informative. I therefore discuss all of the estimates and the picture that they paint even though the results in many of these cases are not precisely estimated enough for more than a suggestive finding.

In the full sample, the reform is associated with a decrease in the wife's resource share of 0.044. In other words, couples in treated states allocate 4.4 percentage points less of their household budget towards the wife. However, the reform also leads to a decrease in the husband's resource share of 0.038. These adult declines correspond to an increase in the share per-child of 0.052 (or an increase of 0.082 for all children). Note that while shares sum to one across household members, the child's share is divided by the number of children in the household. The adult shares require no analogous scaling because every household in the sample consists of exactly one married man and one married woman. While these results do not show a significant treatment effect on the husband's share, we can comfortably exclude the possibility of the husband's share increasing

under this specification. This means that the decrease in the wife's share is entirely absorbed by the increase in the child's share.

With three resource shares being estimated, the underlying changes to bargaining power cannot be read directly off the changes to resource shares. Recall that the household bargaining model does not assign a Pareto weight to children; the observed increase in the child's share therefore cannot be identified as an increase to their bargaining power. The only way the child's share can change as a result of bargaining is if spending on children is demanded by one of the adults. Furthermore, the fact that both adult shares decreased makes it ambiguous which adult overall gained bargaining power. If in contrast the results showed an increase to one adult's share and a decrease to the other's, then we might suspect that the spouse whose share increased was the one whose bargaining power rose. However, even this would depend on how the increase in the child's share is interpreted.

One approach would be to impose some lower bound on the relationship between spouses for spending on children. Previous findings (e.g., Chiappori, Meghir, & Okuyama, 2024) have shown that married women prefer higher spending on children than their husbands do. It is possible to find a set of preferences for each spouse for spending on children that show the reallocation in the full sample as improving the wife's outcome. The preferences would simply have to show that the wife values the increase in the child's share more than she is hurt by the decrease to her own share.

Rather than relying on such assumptions about preferences, I instead inspect the effects of the reform on the four groups in columns 2–5. The purpose is to determine whether there are differences in outcomes across the four groups that are consistent with shifts to bargaining power that are predicted by the LIC model. While only one of the 12 effect sizes is significant, the point estimates all follow the same pattern where the spouse who is the likelier filer experiences a smaller decline in their share than their partner. In the low-probability group, when the husband is the likelier filer the wife's share falls (-0.047) by more than the husband's (-0.025); when the wife is the likelier filer the change in shares for both partners flips such that the wife's share falls less (-0.023) than the husband's (-0.036). This pattern is not only repeated in the two high filer groups, but the gap between each partner's share is also wider than in the two low-filer groups. Finally, in the high probability group where the wife is more likely to file results show that the per-child share increased by a significant 0.081 points. This is consistent with an interpretation that a large increase to a wife's outside

option increases the household's spending on children.

The without children panel in Table 6 shows that the reform did not have an overall an effect on resource shares for these couples. The vast majority of couples without children are sorted into the high probability group where the husband is relatively more likely to file. This makes sense since those without children likely have lower divorce costs both implicit and explicit. The split for couples without children also makes effect sizes for the other three groups too thin to estimate. However even in the high probability group where the husband is relatively more likely to file the effect size is virtually zero.

Table 6: Effect of Reform by Children and Interacted Filing Probability

P(Unilateral Divorce)		Low		High	
Predicted Filer	Full Sample	Husband	Wife	Husband	Wife
With Children					
Wife's Share	-0.044* (0.0251)	-0.047 (0.0387)	-0.023 (0.0325)	-0.079 (0.0656)	-0.038 (0.0379)
Husband's Share	-0.038 (0.0215)	-0.025 (0.0311)	-0.036 (0.0414)	-0.005 (0.0684)	-0.063 (0.0340)
Child's Share	0.052* (0.0207)	0.044 (0.0312)	0.032 (0.0303)	0.074 (0.0985)	0.081* (0.0360)
Without Children					
Wife's Share	-0.006 (0.0337)	0.016 (0.2181)	- -	-0.004 (0.0338)	-0.207 (0.1970)
With Children					
N	42,147	11,450	17,657	1,816	11,224
Control Average					
Wife	0.434	0.406	0.406	0.472	0.468
Husband	0.389	0.378	0.375	0.404	0.402
Child	0.096	0.106	0.105	0.110	0.095
Without Children					
N	16,494	143	-	15,857	423
Control Average	0.568	0.559	0.581	0.568	0.593

Bootstrap standard errors from 960 replicates resampling states, with each replicate used to estimate resource shares with 3SLS and group-time ATTs to accommodate generated regressor as outcome.

The child's share is divided by the number of children in the household.

* 95% percentile bootstrap confidence interval excludes zero.

6.2 Evidence for the LIC model

6.2.1 Direction of Shift in Bargaining Power

Table 6 offers evidence in support of the hypothesis that the spouse more likely to file gained bargaining power, but it also raises several concerns for that interpretation. The supportive finding is that among couples with children, the adult more likely to file increases their share relative to their spouse in all four groups. The change in signs on the husband's own share between the two low-probability groups in particular shows that the identity of the predicted filer matters for who gains from the reform. However, this pattern is not by itself sufficient to establish that the likely filer's bargaining power rose. Two features of the results complicate this reading: (a) the child's share appears to increase across all four groups with children (it does in the fourth group), and (b) both adult shares appear to fall in all four groups (although only one decrease is significant). Additionally, the lack of a similar pattern of changes to resource shares for couples without children suggests that effect is moderated by the presence of children.

The rise in the child's share across all four groups of couples with children is unexpected and complicates interpretation. Both partners plausibly place some value on child spending as a household public good, so both benefit when it rises. However, within the bargaining process only one partner should be driving that increase. In a collective household, the quantity of a public good provided at the Pareto frontier is a weighted average of the two partners' preferences over the household budget. For the child's share to rise as a consequence of a shift in bargaining power, the spouse gaining power must be the one who values child spending relatively more than their partner. Modeled solely as a change in bargaining power, then the consistent increase in the child's share would require that the partner who values child spending more is also the partner relatively more likely to file in every group. If women on average value spending on children more than men do, then this would suggest that women gained bargaining power in each group regardless of who was more likely to file. However, this would not explain the observed decrease to the wife's share in three of the four groups.

A second feature of the results that is hard to reconcile with a pure bargaining shift is the decline in both adult shares in the high likelihood groups. Assuming personal and children's clothing are normal goods, we would expect the spouse whose command over household resources expands should raise both their own

consumption and the child's share in fixed proportions. Even in the low-probability groups where only one spouse's share declines, the increase in the likely filer's personal share is only a fraction of the increase in the child's share. For that to happen, both partners would have to value child spending several times more than their own consumption. One resolution would be to consider children's clothing a luxury good for the spouses. Since bargaining power assigns each spouse a portion of total household consumption, increasing bargaining power for the wife raises her budget; if spending on children is something she values more as her budget increases, then raising her bargaining power may raise her demand for spending on children relative to her own personal consumption. However, if both spouses have these types of preferences then the spouse losing power will demand less spending on children as his budget share goes down. This would be inconsistent with the finding that both adult shares decreased. Additionally, a difference in the expenditure elasticity of demand for the child's share compared to the adult shares would conflict with the share identification strategy. If the child assignable good has systematically higher expenditure elasticity than the adult assignable goods, then the differences in Engel-curve slopes across household members no longer reflect resource shares alone but instead would combine the share with this elasticity gap.

Since the two above patterns cannot be reconciled as shifts to bargaining power, it may be the case that the adoption of NFUD affects intrahousehold resource share allocation through a different component of the collective household problem; the best candidate is individual demand functions. Consider that the reform not only lowers the cost of divorce, but it also has the secondary effect of increasing the overall probability of divorce occurring at a certain time in the future. Even though the marriage participation constraint for a specific couple might be currently satisfied, couples expect that there may be further shifts in each spouse's value in marriage and in divorce in the future. The reduction in the cost of divorce means that small future shifts that would not have significantly mattered previously now have a stronger bite into the marriage constraint. Divorce is therefore evaluated not just as the alternative to staying married, but as something that always has a certain probability of occurring. As a result, as the probability of divorce increases, couples prepare for this possibility by smoothing consumption between the married and divorced states.

Spending on children might be one way to smooth consumption between the married and divorce states. Spending on children is cheaper in marriage due to higher incomes, the public good feature of this spending, and economies of scale. This type of spending has durable value that persists into the divorced state. Early

childhood investment in particular has been shown to be important for later child development outcomes. An increase in the probability that those investments will fall due to divorce creates an added incentive to these investments while the marriage is still intact. Voena (2015) provides evidence for a parallel case showing that when household bargaining power is unequal, a divorce regime with equal property division creates an incentive for increased asset accumulation. This incentive is a self-insurance mechanism since the spouse with more bargaining power must save more to mitigate his drop in consumption if divorce occurs.

One factor specific to this child investment channel is that women are much more likely to receive custody of children than their partner (in 2024, about 75% of divorces with custody agreements resulted in the wife being granted full custody). A direct consequence of this unequal custody outcome is that in the divorced state the wife enjoys more of the value of previous investments in children than the husband does. An increase to the probability of divorce might therefore cause her demand for spending on the child's share to increase while her husband's demand falls. These are however unlikely to be symmetrical effects that cancel each other out. First, if she values the return to this investment in children relatively more than her husband, then the gain to her from this investment should be greater than his loss. Second, even if the husband does not gain custody, he is likely to still place some value on his children's well-being. As the probability of divorce rises, investing more into children during the marriage can therefore be a strategy for the husband to get the most value when the divorce eventually occurs.

Such a child investment channel is consistent with the results in Table 6 when paired together with the main bargaining channel. The increase to the child's share in the two groups where the husband is more likely to file is explained by the wife's, and to an extent her husband's, increased demand for child investment pushing the child's share up. While the household's in-marriage utility for child spending falls when the husband gains bargaining power, the increased probability of divorce raises the returns to child investment. While the husband ends up contributing less than his wife due to his increased bargaining power, both spouses drive an increase to the child's share due to the child investment channel. The observation that the child's share goes up more in the high probability groups is consistent with this evaluation. In these groups, the higher probability of divorce exacerbates the desire for the wife and possibly her husband to increase spending on children while the marriage is intact.

6.2.2 Intensity Margin

Table 6 also offers evidence supporting the hypothesis that the adoption of NFUD affected households based on how likely they previously were to file for AFUD. The reasoning is that renegotiation occurs only when the outside shock is large enough to make a participation constraint bind. Couples with higher predicted filing rates already had less cushioning within their marriage that would allow them to avoid renegotiation over bargaining power (hence their higher divorce rate) and therefore were the most likely to have to engage in bargaining due to the adoption of NFUD. However, the child investment channel proposed in the previous section makes it less clear whether the larger effect sizes in the high probability groups are a result of the intensity margin of reform or the stronger effect of the child investment channel.

Another unexpected finding is that there are significant shifts in shares for both low-probability groups. Significant reallocation among low-probability couples would imply that the shock pushed a constraint to binding for well over half of married couples. This is difficult to imagine for a number of reasons. Were that the case, divorce rates should have risen far more sharply than the observed roughly twofold increase, since not every triggered renegotiation succeeds, and it would require implausibly large inequality in marital gains for the constraint to bind so widely. Similarly, we would expect that the child investment channel only becomes significant when the probability of divorce rises enough that it is something that couples begin to prepare for.

One possible explanation is that spouses overweigh the small probability that their partner would leave, whether taking a threat from their spouse to leave more seriously or preemptively avoiding the possibility of such a threat. The observation that the low-probability groups are affected might provide evidence for a model where divorce is not the relevant threat that is being used. In this case, a change to the outside option might need to be much smaller and would not necessarily result in divorce. However, the most common alternative threat to divorce that has been studied is one where the outside option is to revert to a non-cooperative household where each spouse invests in their own "separate sphere". If this happened, we would expect the child's share to fall as both spouses would allocate less to the public good. Moreover, there would have to be a mechanism by which the adoption of NFUD affected the value of this non-cooperative outside option.

6.3 Pre-Trends and Event-Study Dynamics

Figures 1-6 report event-time aggregations of the group-time ATTs for the three full-sample shares and for the shares that carry the largest heterogeneity results: the wife's share in the high-propensity husband-filer group, the husband's share in the high-propensity wife-filer group, and the children's share in the high-propensity wife-filer group. Since the ENIGH is fielded biennially, each event-time period spans two calendar years; period zero is the first survey conducted with the reform in force, so it captures households observed either in the adoption year itself (for even-year adoptions) or in the following year (for odd-year adoptions). Point estimates are from the non-bootstrapped original sample and the vertical bands are 95% percentile intervals from the state-clustered bootstrap.

All coefficients are reported relative to the period immediately preceding treatment ($g - 1$). Across all six panels the pre-treatment coefficients are small in magnitude relative to the post-treatment effects and are centered on zero, with no single period showing an effect size greater than those for later treated control groups. In the full sample the pre-period point estimates for the wife's, husband's, and children's shares are much smaller than those after treatment. The same pattern holds in the three heterogeneity groups where the pre-treatment estimates remain close to zero. The event studies provide support that there were no existing differential pre-trends between treated and control cohorts.

The post-treatment event dynamics share a common pattern. In each panel the effect appears at event time zero at close to its full magnitude and remains essentially flat over the following three periods with the wife's and husband's full-sample shares settling near -0.04 and the children's share near 0.05 . The single shift shape is consistent with a mechanism where renegotiation is triggered by a single change in the outside option that produces a persistent reallocation, rather than an effect that builds over time. The absence of visible attenuation also matters for the comparison with two-way fixed effects since it bounds how much of the gap can be attributed to dynamic effects alone.

Another visible feature of each figure is the asymmetry between the pre-treatment and post-treatment confidence interval widths with the bootstrap bands being much wider after adoption than before. This is a direct result of the bootstrap procedure including a treatment indicator. Before treatment, treated and not-yet-treated household resources shares are built from the same coefficients, but after treatment the treated

group's shares also included the treatment indicators. This increases the replicate-to-replicate variance of the post-treatment effect sizes.

6.4 Reconciling Magnitudes with Hoehn-Velasco & Penglase (2021b)

The full sample effect sizes reported in this study are considerably larger than those found in Hoehn-Velasco & Penglase (2021b), though less precisely estimated. Several methodological differences contribute to this gap, and it is important to understand each of them since they speak to substantive choices about how to measure bargaining power.

First, this study estimates a three-share system that separately identifies the children's resource share, whereas Hoehn-Velasco & Penglase estimate only two shares (husband and wife). The effect size that they find when estimating this system for couples without children (-0.015 for the wife's share) is more comparable to the findings for the no children subsample in this study. If there is a reallocation toward children that varies across treatment cohorts and event time, the two-share system absorbs this reallocation into the adult shares, potentially attenuating the measured effect on either adult. This may explain why the authors found no significant effects of the reform for couples with children. The three-share system is preferable because it reveals this reallocation directly, and the results in Table 2 confirm that the child's share accounts for a substantial portion of the total reallocation.

Second, the estimation approaches differ. Hoehn-Velasco & Penglase estimate resource shares using nonlinear GMM, which parametrizes resource shares directly as a linear function of covariates within the moment conditions. This is consistent with the model presented in Dunbar, Lewbel, and Pendakur where resource share enter the individual demand function in two different places. Lechene, Pendakur, and Wolf show as a result of this resource shares can be identified linearly from just one of these: the slope of the Engel curve. Therefore, this study estimates linear Engel curves via three-stage least squares and recovers resource shares as ratios of slope coefficients. Both approaches instrument total expenditure with income to address measurement error, but the functional form assumptions slightly differ.

Third, this study employs the Callaway & Sant'Anna (2021) group-time ATT estimator rather than two-way fixed effects whereas Hoehn-Velasco & Penglase situate a two way fixed effects estimator in their GMM. In a staggered adoption setting where treatment effects vary across cohorts and over time, TWFE can pro-

duce biased estimates through forbidden comparisons identified in Goodman-Bacon (2021). Specifically, when treatment effects attenuate over time, already-treated units serve as inappropriate comparisons for newly-treated units. The attenuation gets combined with the treatment effect itself, with the latter being over estimated. This may explain the significant results that the authors found for childless couples that are not replicated in this paper.

Finally, the two studies differ in their definition of treatment timing. This study defines a state as treated when it is either first legally available or when no-fault unilateral divorce becomes the most common form of unilateral divorce filing. This identification using two different channels allows for the nuance present in the decentralized administration of family law in Mexico.

7 Conclusion

In this paper I examined how lowering the cost of divorce changed the allocation of resources and bargaining power within marriages that continue using Mexico's staggered adoption of no-fault unilateral divorce. I use the limited-commitment intertemporal collective model to derive testable predictions about which households renegotiate as a result of the reform and in which spouse's favor. I estimate resource shares of wives, husbands, and children from assignable clothing Engel curves and estimate reform effects with the Callaway & Sant'Anna (2021) group-time estimator within a state-clustered bootstrap.

This paper has three main findings. First, the reform reallocated household spending toward children such that the per-child share rose by 5.2 percentage points (on the full sample) while both adult shares fell (with the effect being absent among couples without children). Second, the direction of the adult reallocation follows the predicted filer. In every group of couples with children, the spouse more likely to file for unilateral divorce experienced the smaller decline in their own share. Third, both the gap in adult shares and the increase in the children's share are larger among couples with a higher predicted propensity to divorce which the model identifies as the households most likely to engage in bargaining. Though most of the effect size estimates are imprecise on their own, taken together this pattern across 12 estimates is unlikely to occur by chance and provides support for the LIC model.

The findings suggest an additional channel driving reallocation of resource shares in addition to bargain-

ing power. The relative movement of the two adult shares suggests a bargaining channel where an adjustment to Pareto weights is triggered by the reform increasing the value of the outside option. However, the consistent shift toward children that is common to all groups with children and largest where divorce became most likely cannot be explained by bargaining alone. It is instead consistent with an investment response where as a result of an increase in the probability of divorce spending on children becomes more attractive. This would particularly affect wives due to largely expected custody outcomes. Based on this, the reform shifted both the distribution of power between spouses and the household's preference for investing in children.

These results reframe evidence presented in Hoehn-Velasco & Penglase (2021b). While in that study the authors estimated a two-share system finding a modest decline in women's shares, recovering the children's share directly shows that the decline in the wife's share is due to a reallocation toward children rather than toward husbands. This suggests that in households with children, a two-share system can obscure a reallocation directed towards children as a loss for one of the adults.

The evidence comes with limits to the precision of estimates. Bootstrapped standard errors results in group effects that are suggestive. Additionally, the predicted-filer classification is a very rough proxy of actual couple-level changes to outside options. Resource shares are informative to allocation of the household budget, but rest on a strong similar across people (SAP) assumption on preferences.

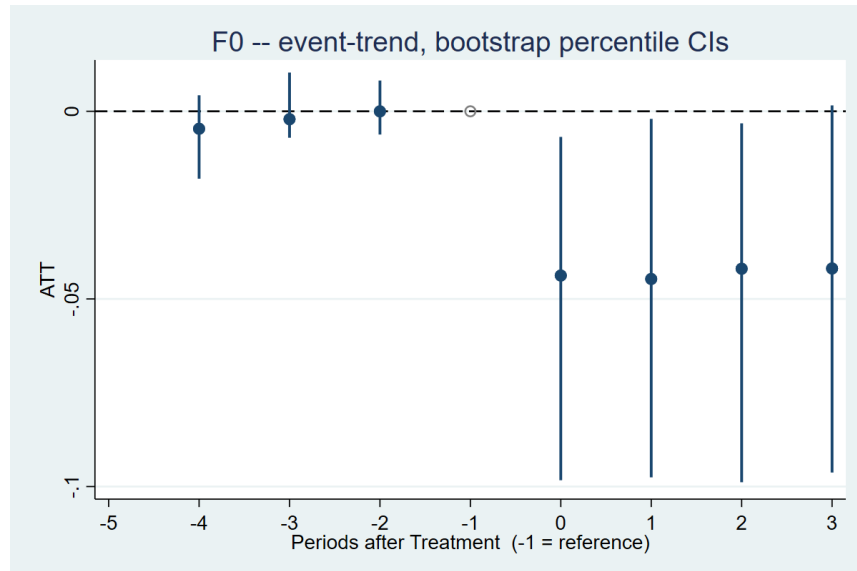


Figure 1: Wife's Share with Children without Heterogeneous Treatment Groups

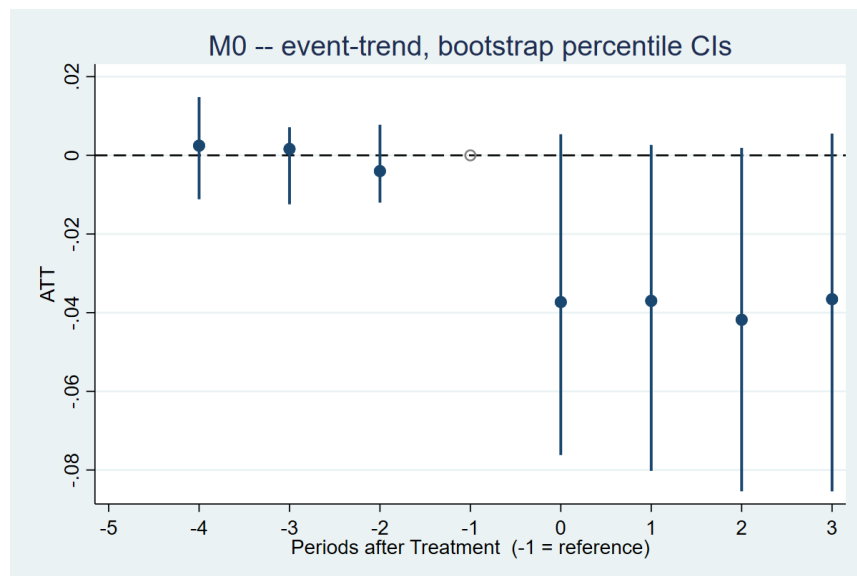


Figure 2: Husband's Share with Children without Heterogeneous Treatment Groups

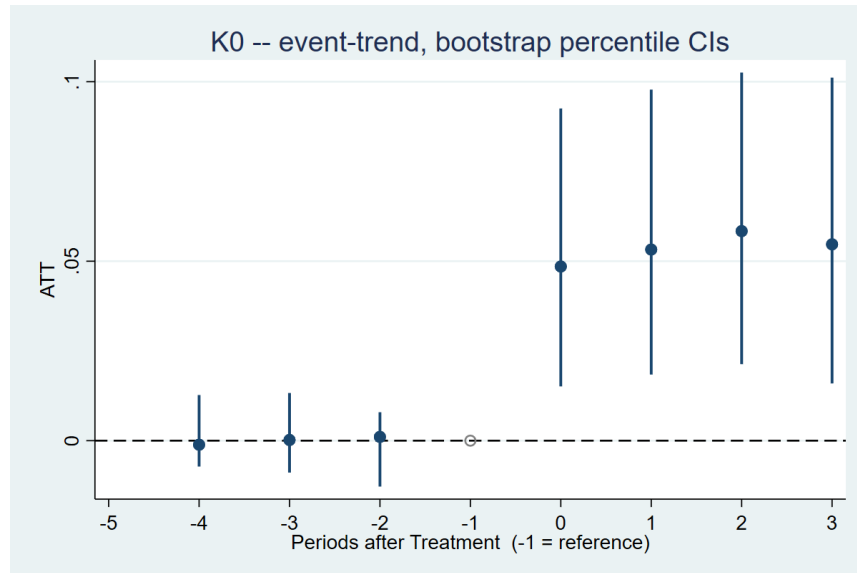


Figure 3: Child's Share without Heterogeneous Treatment Groups

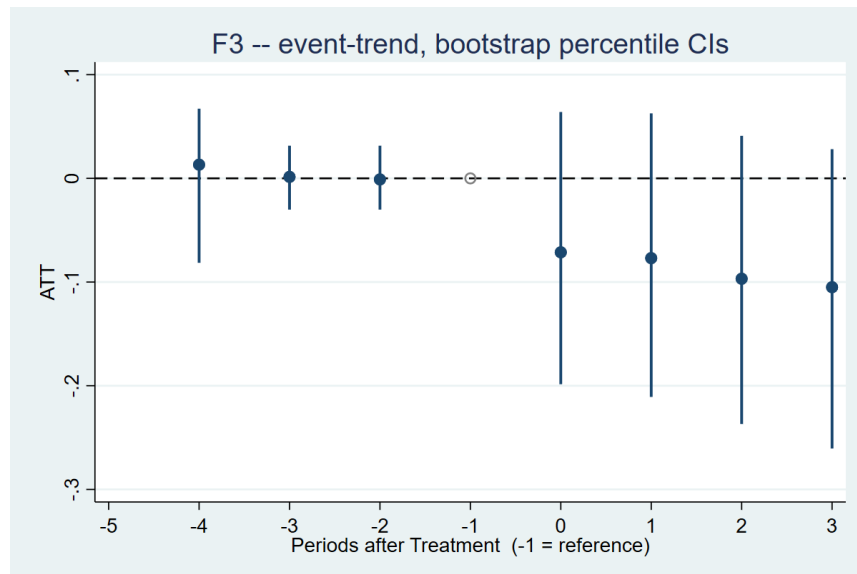


Figure 4: Wife's Share with Children in High Propensity Husband Filer Group

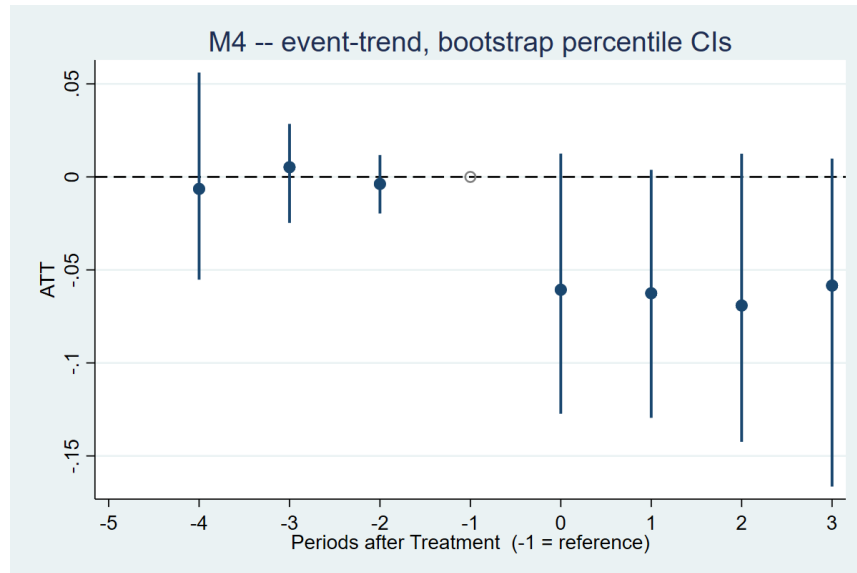


Figure 5: Husband's Share with Children in High Propensity Wife Filer Group

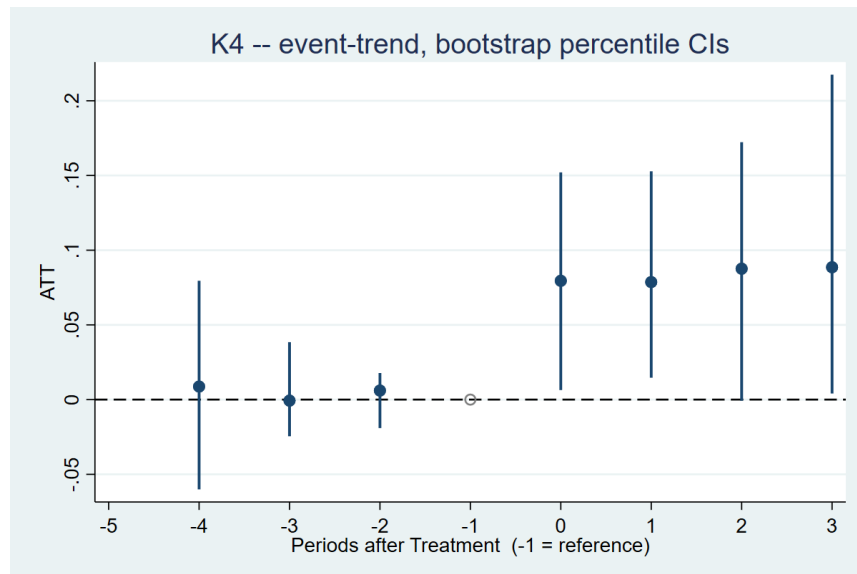


Figure 6: Child's Share in High Propensity Husband Filer Group

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